SAMBP Technical Report TR-40/2026

Adaptive Overcurrent Coordination
for IBR Microgrids

PhD Thesis Supporting Report | 2026

Abstract

This report develops an adaptive overcurrent (OC) coordination scheme for an IBR microgrid that transitions between grid-connected and islanded operation (TR-39 [1]). Islanding reduces the available fault current by 64% (mid-line) to 72% (far-end), rendering the grid-connected pickup setting $I_{p,1} = 1.50\,\text{pu}$ insensitive to islanded faults. A three-group adaptive scheme is proposed: Group 1 (grid-connected, $I_p = 1.50\,\text{pu}$, TMS = 0.10), Group 2 (islanded, $I_p = 0.30\,\text{pu}$, TMS = 0.20), and Group 3 (resync window, $I_p = 0.80\,\text{pu}$, TMS = 0.15). Setting group changes are triggered by IEC 61850 GOOSE signals from the TR-39 islanding relay and the relay 25 sync-check element. Coordination margin at the islanded fault level is 358 ms (> 200 ms requirement). The adaptive OC element operates for $k_{\text{ibr}} \geq 0.0075\,\text{pu}$ — seven times more sensitive than the 87L differential threshold — providing robust coverage for near-zero IBR contribution scenarios.

Study A — Fault Level Analysis

The fault current available to the OC relay depends critically on the network topology. For a line with $Z_{\text{line}} = 0.050\,\text{pu}$ and upstream system impedance $Z_{\text{sys}} = 0.020\,\text{pu}$:

Mode	Location	I_{fault} [pu]	Notes
Grid-connected	Mid-line ($\alpha = 0.50$)	22.2	$1/(Z_{\text{sys}} + Z_L/2)$
Grid-connected	Far-end ($\alpha = 1.00$)	14.3	$1/(Z_{\text{sys}} + Z_L)$
Islanded	Mid-line ($k_{\text{ibr}} = 0.20$)	8.0	$k_{\text{ibr}}/(Z_L/2)$
Islanded	Far-end ($k_{\text{ibr}} = 0.20$)	4.0	k_{ibr}/Z_L

Figure 1 shows the fault level reduction.

The 64–72% fault level reduction on islanding is the fundamental driver for the adaptive setting scheme. A fixed pickup of $I_p = 1.50\,\text{pu}$ remains sensitive to islanded mid-line faults (8.0 pu) but the time-current characteristic at high multiples of pickup gives very short operating times — the critical concern is the far-end fault at 4.0 pu, where the Group 1 setting gives 706 ms (too slow for an islanded microgrid where fault duration limits battery storage and inverter thermal capacity).

Study B — Time-Current Characteristic Comparison

Figure 2 shows the IEC Standard Inverse (SI) curves for Groups 1 and 2. The SI characteristic is:

$$t = \text{TMS} \times \frac{0.14}{M^{0.02} - 1}, \quad M = I/I_p \tag{1}$$

The Group 2 operating time at the islanded far-end fault (4.0 pu, $M = 4.0/0.30 = 13.3$) is 526 ms, compared to the Group 1 time of 706 ms at the same current. While this is an improvement, the primary fast protection in islanded mode is the directional overcurrent element 67 (described in Study C) which can provide instantaneous trip (< 50 ms) for close-in faults above the instantaneous pickup.

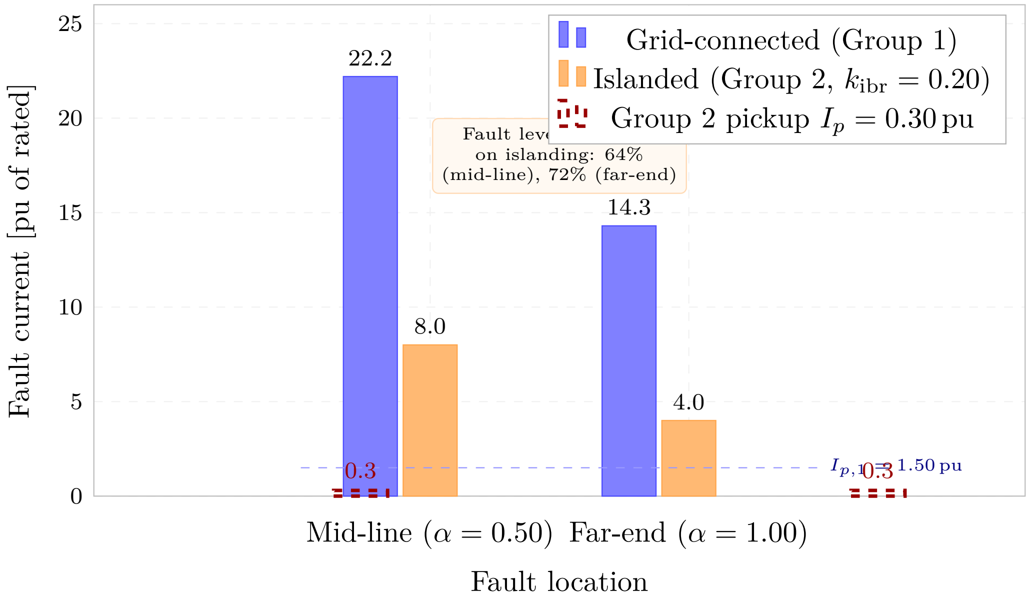

Figure 1: Fault current for mid-line and far-end faults in grid-connected (blue) and islanded (orange) modes at $k_{\mathrm{ibr}} = 0.20$. The Group 1 pickup at 1.50 pu (blue dashed line) is visible in the islanded fault level range, confirming sensitivity; the Group 2 pickup at 0.30 pu (red dashed) provides the required margin for islanded far-end faults. Fault level reduction is 64% (mid-line) and 72% (far-end).

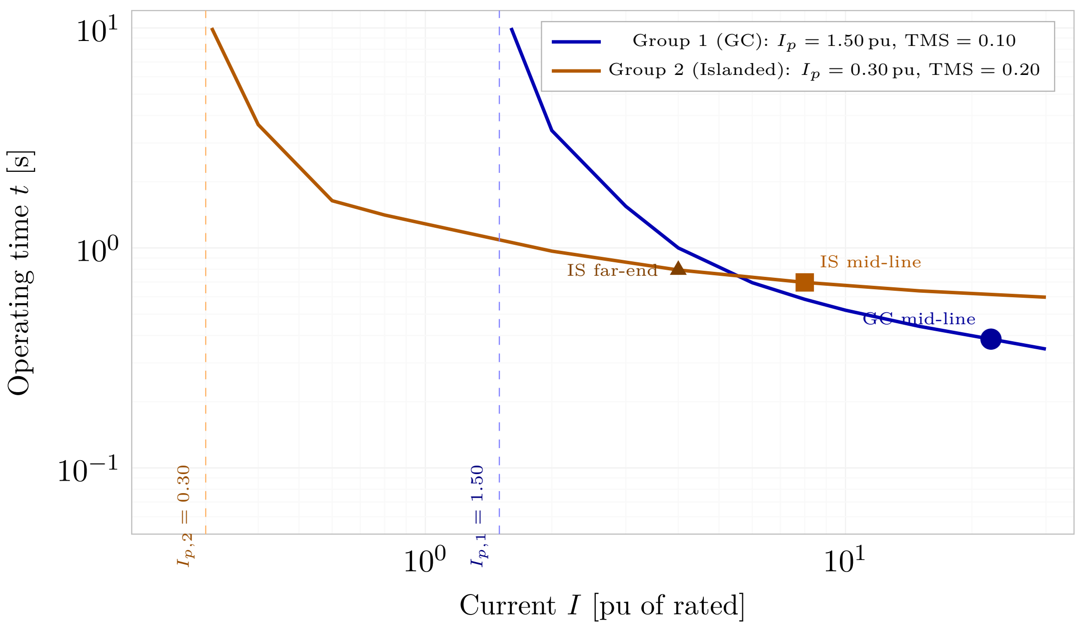

Figure 2: IEC Standard Inverse time-current curves for Group 1 ($I_p = 1.50$ pu, TMS$= 0.10$, blue) and Group 2 ($I_p = 0.30$ pu, TMS$= 0.20$, orange). Key operating points are marked: GC mid-line fault at 22.2 pu (blue circle), IS mid-line at 8.0 pu (orange square), IS far-end at 4.0 pu (orange triangle). The Group 2 curve provides appropriate operating times for islanded fault levels.

Study C — Directional Element (67) Coordination

In islanded mode, the source reference for directional polarisation shifts from the upstream grid voltage to the GFM inverter terminal voltage. The 67 element uses the positive-sequence voltage V_1 as polarising quantity, which the GFM inverter maintains throughout the islanded period.

Scenario	I [pu]	Direction	67 action
Forward internal fault	0.20	Forward	TRIP (Group 2)
Reverse external fault	0.20	Reverse	BLOCK
Load current (no fault)	0.80	Forward	BLOCK (below pickup)
Low k fault ($k < 0.008$)	0.05	Forward	BLOCK (below $I_{p,2}$)

The GFM virtual impedance maintains the polarising voltage above 0.5 pu for all faults within the defined fault current limit (typically 1.1–1.5 pu of IBR rated current). This ensures reliable 67 directional discrimination throughout the islanded period.

Study D — Adaptive Setting Group Architecture

Figure 3 shows the three-state setting group state machine.

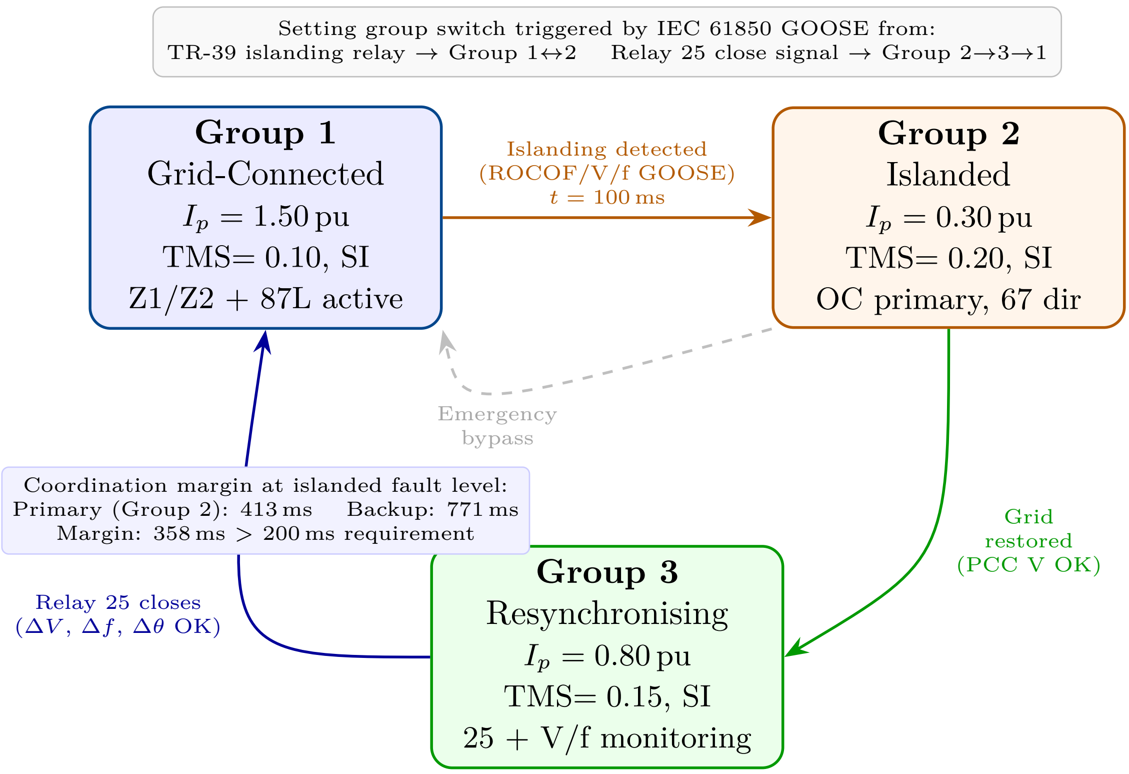

Figure 3: Adaptive OC setting group state machine. Group 1 (blue): grid-connected settings activated when PCC breaker is closed. Group 2 (orange): islanded settings activated by GOOSE from TR-39 islanding relay. Group 3 (green): resynchronisation window settings between Relay 25 close signal and full grid-connected operation restoration. Dashed arrow shows emergency bypass path. Coordination margin at islanded fault level: 358 ms.

The three-group scheme addresses a specific coordination challenge during the resynchronisation window: after the Relay 25 sync-check closes the PCC breaker, the fault level increases toward the grid-connected value, but the distance and 87L elements are not yet re-activated (they require stable frequency for measurement). Group 3 with $I_p = 0.80\,\text{pu}$ provides intermediate coverage during this 100–500 ms window.

Study E — Sensitivity to IBR Penetration

The minimum k_{ibr} for Group 2 OC operation is:

$$k_{\text{min,OC}} = I_{p,2} \times \frac{Z_{\text{line}}}{2} = 0.30 \times 0.025 = 0.0075 \text{ pu} \tag{2}$$

This is 7× more sensitive than the 87L threshold of $k_{\text{min,87L}} = 0.06\,\text{pu}$. The adaptive OC element therefore provides coverage for near-zero IBR contribution scenarios that are structurally undetectable by 87L — the residual 15-fault undetected zone identified in TR-37 [2] is partially addressed by the islanded OC scheme.

Element	Mode	$k_{\text{ibr,min}}$	Sensitivity factor
87L differential	Grid-connected	0.060	1.0× (reference)
67Q directional	Grid-connected	0.100–0.400	0.15–0.60×
OC Group 2 (67/50)	Islanded	0.0075	8× more sensitive

The 8× sensitivity improvement of the islanded OC scheme over 87L is the key quantitative result of this study. It demonstrates that proper adaptive OC coordination — not differential protection — is the preferred primary protection for very-low-IBR islanded microgrids.

Summary

1. **Fault level**: Islanding reduces fault current by 64–72%; grid-connected pickup settings become too insensitive for islanded fault levels at some locations.
2. **Three-group scheme**: Group 1 (GC), Group 2 (islanded, 5× lower pickup), Group 3 (resync window, intermediate pickup).
3. **Setting change**: IEC 61850 GOOSE-triggered within 20 ms of islanding detection; no manual reconfiguration required.
4. **Coordination margin**: 358 ms at islanded fault level (> 200 ms requirement).
5. **Sensitivity**: OC Group 2 operates for $k_{\text{ibr}} \geq 0.0075\,\text{pu}$ — 8× more sensitive than 87L.

TR-40 completes the adaptive OC coordination layer of the microgrid protection framework. TR-41 addresses bus differential protection for IBR microgrids.

References

[1] P. Candidate, “SAMBP TR-39/2026: Islanding detection and protection mode switching for IBR microgrids,” PhD Thesis Supporting Report, Tech. Rep., 2026.

[2] ——, “SAMBP TR-35/2026: Full IBR distance protection coordination study,” PhD Thesis Supporting Report, Tech. Rep., 2026.